

BHAVENTHAN R

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SUMMARY

Final-year Electronics and Communication Engineering student with hands-on experience in Java backend development, JDBC, and MySQL-based systems. Built multiple academic projects implementing CRUD operations, backend business logic, and hardware–software integration. Strong foundation in Data Structures & Algorithms, Object-Oriented Programming, and SQL. Actively solving DSA problems and seeking entry-level Java Developer / Software Engineer roles.

EDUCATION

B.E. Electronics and Communication Engineering (2022 - 2026) | **CGPA: 8.44**

Sathyabama Institute of Science and Technology, Chennai

HSE (12th), State Board Examination, Tamil Nadu (2022) | **75.1%**

SSLC (10th), State Board Examination, Tamil Nadu (2020) | **99.2%**

SKILLS

Programming Languages: Java, SQL, Python (basics)

Core Java: OOPs, Collections Framework, Exception Handling, Multithreading (basics)

Core CS: Data Structures & Algorithms, OOPs, Computer Networks

Frameworks/Tools: Git, GitHub, JDBC, Maven, Postman

Database Management: MySQL

Productivity Tools: MS Excel, MS Word, PowerPoint

Languages: Tamil, English, German (basics)

PROJECTS

- **Attendance Management System | Java + MySQL**

Built a Java-based attendance management system using JDBC and MySQL to manage student records, implement CRUD operations, and export reports to Excel using Apache POI.

Tech Stack: Java, JDBC, MySQL, Apache POI

- **Vibration-Based Predictive Maintenance for Motors | Raspberry Pi + Edge AI**

Designed a vibration-based monitoring system to predict early motor failures by analyzing sensor data patterns and applying machine learning concepts to reduce industrial downtime, Published in IEEE conference.

Tech Stack: Raspberry Pi, ADXL345, Machine Learning

- **Home Security System | ESP32 (IoT)**

Created an IoT-based security system using ESP32 to detect intrusions and send real-time alerts, coordinating hardware integration and alert logic through collaborative testing.

Tech Stack: ESP32, Sensors, IoT, MQTT Broker, Mobile Alerts, Home Automation

- **Smart Belt for Coal Miners | ESP32 Microcontroller (IoT)**

Designed an IoT-based smart belt using ESP32 to monitor miners' safety parameters with real-time alerts; the work was published in **JCBECA** Journal

Tech Stack: ESP32, Sensors, IoT, MQTT Broker

CERTIFICATIONS & ACHIEVEMENTS

- Summer School Program on “Semiconductor Technology” - IISc Bangalore
- NPTEL: Python for Data Science (4-week course)
- Solved 200+ DSA problems using Java and Python in **Leetcode** - leetcode.com/u/bhaventhan
- Presented and participated in an IEEE conference at SRM Institute of Science and Technology, showcasing a research paper in the field of **Embedded Systems and Intelligent Technologies**.
- IEEE Published Author — “Vibration-Based Predictive Maintenance for motors using Edge AI” (ieeexplore.ieee.org/document/11504862)